

REVISITING

KAPLANSKY'S FIFTH CONJECTURE

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UNIVERSIDAD
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SPAIN

Funded by:



Outline

- 1 Kaplansky's fifth conjecture
- 2 Wedderburn blocks of size at most two
- 3 The pointed case



1. Kaplansky's fifth conjecture

Introduction



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University of Chicago, 1975.



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1917-2006

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Y. Sommerhäuser, *On Kaplansky's conjectures*, 393-412. Lecture Notes in Pure and Appl. Math. 210. Dekker, 2000.

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- ❖ If, in addition, H is cosemisimple, then $S^4 = \text{id}$.

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Ring of Witt vectors. Every semisimple and cosemisimple Hopf algebra lifts to characteristic zero.

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Usual assumption. $\text{char}(k) \nmid 2 \dim(H)$:

$$S^{4 \dim(H)} = \text{id} = (S^2)^{2 \dim(H)} \Rightarrow S^2 \text{ diagonalizable.}$$

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Proposition (Speaker). *Suppose that $\text{char}(k) = 2$ and $n_i \leq 2$ for all i . Then, S^2 is diagonalizable.*

Proof. ① Suppose that $He_i \cong k$. Then, $S^2(e_i) = e_i$.

② Suppose that $He_i \cong M_2(k)$. Let $\lambda : H \rightarrow k$ be a left integral. There is a unique $u \in He_i$ invertible such that $\lambda(x) = \text{Tr}(ux)$ for all $x \in He_i$. Moreover, $S^2|_{He_i}$ is conjugation by u and $\text{Tr}(S^2|_{He_i}) = \lambda(e_i) = \text{Tr}(u) \neq 0$. Let J be the Jordan canonical form of u . If u had only one eigenvalue, then J would be a scalar matrix or a Jordan block. Hence, $\text{Tr}(u) = 0$ ($\text{char}(k) = 2$), a contradiction.

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Corollary. *Suppose that $\text{char}(k) = 2$ and $n_i \leq 2$ for all i . Then, H is commutative.*

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A new result

Theorem (*Vendramin-Speaker*). *Let H be a pointed and semisimple Hopf algebra over a field of positive characteristic p .*

(1) *If $p \neq 2$, then $S^2 = \text{id}$.*

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- !! Lift this property through the coradical filtration using the Taft-Wilson Lemma.

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Proof: Step 1



S. Scherotzke, *Classification of pointed rank one Hopf algebras*. J. Algebra 319 (2008), 2889-2912.

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$$\Delta(x) = 1 \otimes x + x \otimes g, \quad S(x) = -xg^{-1}, \quad S^2(x) = gxg^{-1}.$$

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$$d = Np^r, \quad y^{Np^r} = \alpha_{-1}(g^{Np^r} - 1) + \sum_{i=0}^{r-1} \alpha_i y^{Np^i}, \quad \alpha_i \in k.$$

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A basis of B is

$$\{g^\alpha y^\beta : 0 \leq \alpha < n, 0 \leq \beta < Np^r\}.$$

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Hence, $\omega = 1$, and the map

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is the identity.

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$$c = \sum_{g,h \in G(C)} c_{g,h}, \quad \text{where} \quad \Delta(c_{g,h}) = g \otimes c_{g,h} + c_{g,h} \otimes h + w_{g,h},$$

for some $w_{g,h} \in C_{n-1} \otimes C_{n-1}$.

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For $g, h \in G(C)$, consider the subspace

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Proof. We proceed by induction on n . Take $n \geq 2$.

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$$\Delta(f(c) - c) = g \otimes (f(c) - c) + (f(c) - c) \otimes h$$

and $f(c) - c \in C_1$.

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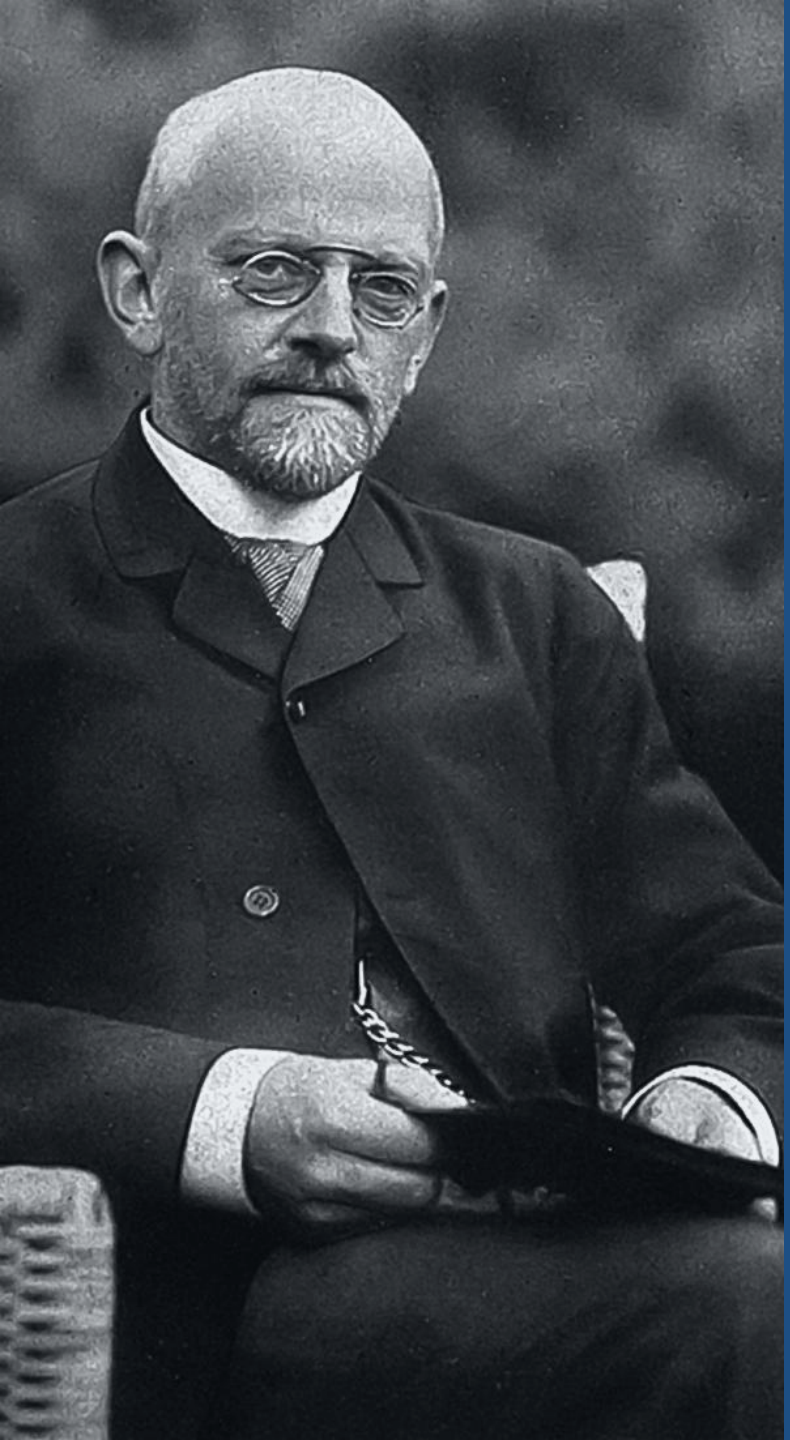
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1862-1943